Homework 2 (LP Formulation and Conversion to Standard Form)

January 27, 2026

This assignment is focused on modeling LPs. It is due on Thursday, Jan. 29th. at Midnight. Questions 1 and 2 will be worth 10 points. Questions 3 and 4 will be worth 20 points. You must show your work to receive credit. You are encouraged to work with each other, but you must write up and submit your own assignment.

1)

Show that any LP in standard form can be converted to the form:

$$\text{minimize } z = \vec{c} \cdot \vec{x}$$

subject to

$$A\vec{x} \geq \vec{b}$$

$$\vec{x} \geq 0$$

(Note there are no equality constraints in this form. Only inequality constraints)

2)

Convert the following LP into standard form:

$$\text{minimize } z = |x_1| + |x_2 + x_3|$$

s.t.

$$x_1 - x_2 \leq x_3 + 4$$

$$x_1 \geq 0$$

$$x_2 \leq 0$$

$$x_3 \in \mathbb{R}$$

3)

a)

Formulate the following problem into an LP. Be sure to explain the units for each decision variable.

Lynn is able to knit scarves and hats. Each scarf takes 200 yards of red yarn and 300 yards of blue yarn. Each hat takes 50 yards of red yarn and 400 yards of blue yarn. Lynn is able to sell the hats for \$20 each and the scarves for \$25 each. If Lynn has 1000 yards of red yarn and 2800 yards of blue yarn, and she wishes to maximize how much money she makes from selling knitted goods, find the optimal number of each product Lynn should knit.

b)

Solve the LP from **a)**.

4)

a)

Consider an apiary with I old beehives, J new beehives, and K types of bees. Each new beehive j has a capacity of C_{jk} bees of type k. Each old beehive has O_{ik} bees of type k. The distance between old beehive i and new beehive j is d_{ij}. Formulate an LP whose objective is to minimize the total distance traveled by all of the bees, if each bee is relocated from their old beehive to exactly one new beehive. (You can ignore the assumption that the number of bees must be an integer).

b)

Explain each constraint in the model. Explain what your decision variables are and be certain to specify the units.

c)

Convert your model to standard form.